Business Case: Netflix – Data Exploration and Visualization

STAGE 1 — DATA EXPLORATION

1. Business Understanding

Netflix wants to understand:

Which type of content should they produce?

Which countries should they focus on?

Which genres are most popular?

What is the best time to release content?

Which markets have the highest growth potential?

Goal: Converting raw Netflix data into business insights to support content acquisition and expansion strategies.

2. Import Libraries

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use("ggplot")

pd.set_option("display.max_columns",None)
```

3. Load Dataset

```
!pip install gdown
```

```
Requirement already satisfied: gdown in /usr/local/lib/python3.12/dist-packages (5.2.2)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-packages (from gdown) (4.13.5)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from gdown) (3.29.4)
Requirement already satisfied: requests[socks] in /usr/local/lib/python3.12/dist-packages (from gdown) (2.32.4)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from gdown) (4.67.3)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->gdown) (2.8.4)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->gdown)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown) (3.18)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown) (2.5.
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown) (2026
Requirement already satisfied: PySocks!=1.5.7,>=1.5.6 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown) (
```

```
import gdown
netflix = "1piPkFMfP1Oj5JbR9HIHic8kt6hmzGDUs"
gdown.download(
    f"https://drive.google.com/uc?id={netflix}",
    "netflix.csv",
    quiet=False
)
df = pd.read_csv("netflix.csv")
df.head()
```

```
Downloading...
From: https://drive.google.com/uc?id=1piPkFMfP1Oj5JbR9HIHic8kt6hmzGDUs
To: /content/netflix.csv
100%|██████████| 3.40M/3.40M [00:00<00:00, 25.6MB/s]
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV	Blood &	NaN	Ama Qamata, Khosi Ngema,	South	September	2021	TV-MA	2	International TV Shows, TV	After crossing paths at a

Next steps: Generate code with df New interactive sheet

4. Basic Information:

> Is the dataset complete and suitable for analysis?

```
df.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV	Blood &	NaN	Ama Qamata, Khosi Ngema,	South	September	2021	TV-MA	2	International TV Shows, TV	After crossing paths at a

Next steps: Generate code with df New interactive sheet

```
df.tail()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	descriptio
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007	R	158 min	Cult Movies, Dramas, Thrillers	A politic cartoonist, crim reporter ar a
											Kids' TV	While livir

```
df.sample(5)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	descripti
4788	s4789	Movie	Katt Williams: Kattpacalypse	Marcus Raboy	Katt Williams	United States	July 3, 2018	2012	TV-MA	61 min	Stand-Up Comedy	Urban cor Katt Willia ushers Kattpacal
6053	s6054	Movie	A Princess for	Michael	Katie McGrath, Sir Roger Moore,	United	August 28,	2011	TV-PG	91 min	Children & Family Movies,	At invitation c relati

```
df.shape
```

```
(8807, 12)
```

Observation

> Dataset contains 8807 rows and 12 columns.

5. Data Types

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

Observation

> Most categorical columns are stored as object datatype.

6. Duplicate Check

```
df.duplicated().sum()
```

```
np.int64(0)
```

Observation

> No duplicate records found.

7. Missing Values

```
df.isnull().sum()

(
df.isnull()
.sum()
.sort_values(ascending=False)
)
```

	0
director	2634
country	831
cast	825
date_added	10
rating	4
duration	3
show_id	0
type	0
title	0
release_year	0
listed_in	0
description	0

dtype: int64

Observation

> Director, Cast and Country contain the highest number of missing values.

Business implication

> Missing director and cast information may affect actor/director trend analysis.

8. Summary Statistics

```
# Numerical

df.describe()
```

	release_year
count	8807.000000
mean	2014.180198
std	8.819312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

```
# Categorical

df.describe(include="all")
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	descripti
count	8807	8807	8807	6173	7982	7976	8797	8807.000000	8803	8804	8807	88
unique	8807	2	8807	4528	7692	748	1767	NaN	17	220	514	87
top	s8807	Movie	Zubaan	Rajiv Chilaka	David Attenborough	United States	January 1, 2020	NaN	TV-MA	1 Season	Dramas, International Movies	Paranorm activity a lus abandon prope
freq	1	6131	1	19	19	2818	109	NaN	3207	1793	362	
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2014.180198	NaN	NaN	NaN	Na
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.819312	NaN	NaN	NaN	Na
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1925.000000	NaN	NaN	NaN	Na
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2013.000000	NaN	NaN	NaN	Na
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2017.000000	NaN	NaN	NaN	Na

9. Unique Values

```
df.nunique()
```

	0
show_id	8807
type	2
title	8807
director	4528
cast	7692
country	748
date_added	1767
release_year	74
rating	17
duration	220
listed_in	514
description	8775

dtype: int64

```
for col in df.columns:
    print(col)
    print(df[col].nunique())
```

```
show_id
8807
type
2
title
8807
director
4528
cast
7692
country
748
date_added
1767
release_year
74
rating
17
duration
220
listed_in
```

```
514
description
8775
```

Business implication

> Netflix offers multiple ratings and genres to serve diverse audience groups.

10. Dataset Memory

```
df.memory_usage(deep=True)
```

	0
Index	132
show_id	474471
type	480930
title	599158
director	495687
cast	1631213
country	517757
date_added	561028
release_year	70456
rating	470514
duration	493486
listed_in	725748
description	2065368

dtype: int64

Business implication

> Converting categorical variables reduces memory usage.

Double-click (or enter) to edit

STAGE 2 — DATA-CLEANING

```
## Business Question

# Can we improve the quality of the data before analysis?


### Missing Values

df['director']=df['director'].fillna("Unknown")

df['cast']=df['cast'].fillna("Unknown")

df['country']=df['country'].fillna("Unknown")

# Drop

df.dropna(subset=['date_added'],inplace=True)
```

Observation

Critical missing records removed

```
# Convert Date

df['date_added']=pd.to_datetime(df['date_added'],format="mixed")


# Extract

df['added_year']=df['date_added'].dt.year

df['added_month']=df['date_added'].dt.month

df['added_month_name']=df['date_added'].dt.month_name()

df['added_day']=df['date_added'].dt.day

df['added_weekday']=df['date_added'].dt.day_name()
```

```
df.head(15)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	descri
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her nea end life, f
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	cr pat party, a To
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To prot family po dru
3	s4	TV Show	Jailbirds New Orleans	Unknown	Unknown	Unknown	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	flirtatio toilet down
4	s5	TV Show	Kota Factory	Unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a co c kn t
5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	Unknown	2021-09-24	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries	The ar a chari young
6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	Unknown	2021-09-24	2021	PG	91 min	Children & Family Movies	Equ divide a brigh her

Next steps: Generate code with df New interactive sheet

```
df.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	0
cast	0
country	0
date_added	0
release_year	0
rating	4
duration	3
listed_in	0
description	0
added_year	0
added_month	0
added_month_name	0
added_day	0
added_weekday	0

dtype: int64

Business value

Now Netflix can analyze

Best year Best month Best weekday

```
# Split Multi-value Columns


# Country

country_df=df.copy()
country_df['country']=country_df['country'].str.split(',')
country_df=country_df.explode('country')
country_df['country']=country_df['country'].str.strip()
country_df=country_df.reset_index(drop=True)


# Cast

cast_df = df.copy()
# Split multiple actors
cast_df['cast'] = cast_df['cast'].str.split(',')
# Convert each actor into a separate row
cast_df = cast_df.explode('cast')
# Remove extra spaces
cast_df['cast'] = cast_df['cast'].str.strip()
# Reset index
cast_df = cast_df.reset_index(drop=True)




#Listed_in

genre_df = df.copy()
# Split multiple genres
genre_df['listed_in'] = genre_df['listed_in'].str.split(',')
# Convert each genre into a separate row
genre_df = genre_df.explode('listed_in')
# Remove extra spaces
```

```
genre_df['listed_in'] = genre_df['listed_in'].str.strip()
# Reset index
genre_df = genre_df.reset_index(drop=True)
```

```
country_df.head(10)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	descriptic
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her fath nears th end of h life, filmm
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	Aft crossir paths at party, a Cap Town t
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect h family from powerf drug lor
3	s4	TV Show	Jailbirds New Orleans	Unknown	Unknown	Unknown	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feud flirtations ar toilet talk g down amo
					Mayur More,						International	In a city coachir

Next steps: Generate code with country_df New interactive sheet

```
cast_df[ 'cast'].head(10)
```

	cast
0	Unknown
1	Ama Qamata
2	Khosi Ngema
3	Gail Mabalane
4	Thabang Molaba
5	Dillon Windvogel
6	Natasha Thahane
7	Arno Greeff
8	Xolile Tshabalala
9	Getmore Sithole

dtype: object

```
genre_df['listed_in'].head(10)
```

	listed_in
0	Documentaries
1	International TV Shows
2	TV Dramas
3	TV Mysteries
4	Crime TV Shows
5	International TV Shows
6	TV Action & Adventure
7	Docuseries
8	Reality TV
9	International TV Shows

dtype: object

```
print(df.shape)
print(country_df.shape)
```

```
(8797, 17)
(10840, 17)
```

Each country becomes one observation.

```
# Movie Duration


movies=df[df['type']=="Movie"].copy()
movies=movies.dropna(subset=['duration'])
movies['duration']=movies['duration'].str.replace(" min","")
movies['duration']=movies['duration'].astype(int)
```

```
df['duration'].head(10)
```

	duration
0	90 min
1	2 Seasons
2	1 Season
3	1 Season
4	2 Seasons
5	1 Season
6	91 min
7	125 min
8	9 Seasons
9	104 min

dtype: object

Conclusion:

The movie duration column is successfully cleaned and converted into a numerical format, making it suitable for statistical analysis and visualization.

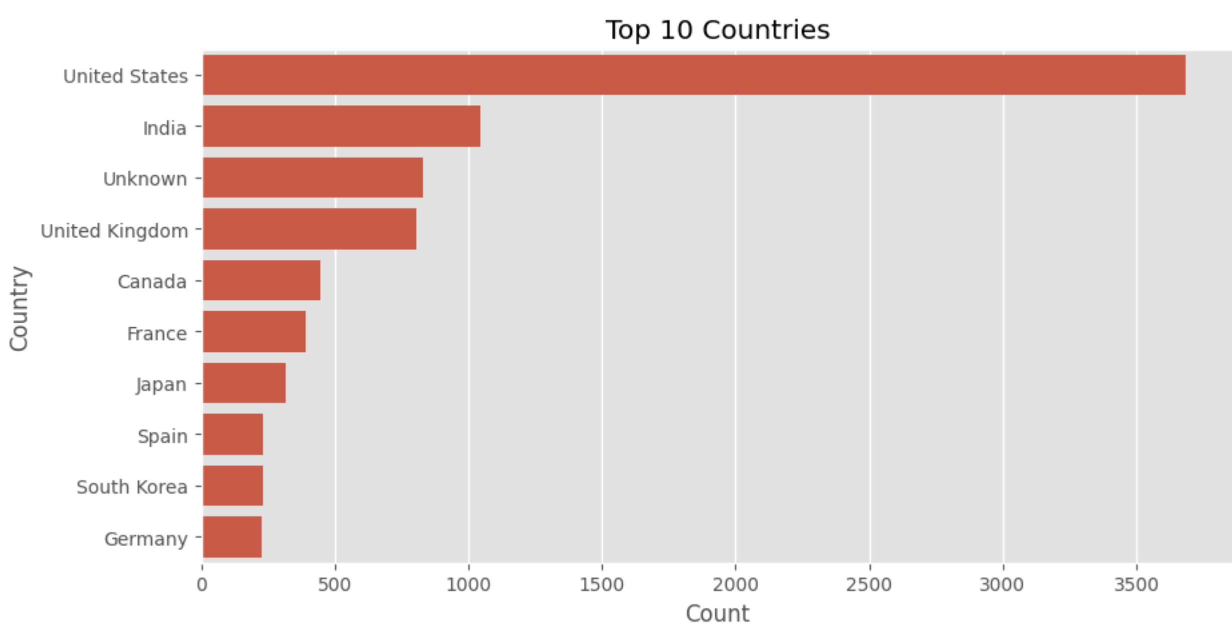
Double-click (or enter) to edit

STAGE 3 —VISUALIZATION

Step 1: Top 10 Countries Code

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
country_df = (
    df.assign(country=df['country'].fillna('Unknown').str.split(', '))
      .explode('country')
      .reset_index(drop=True)
)

plt.figure(figsize=(10,5))

sns.countplot(
    data=country_df,
    y='country',
    order=country_df['country'].value_counts().head(10).index
)

plt.title("Top 10 Countries")
plt.xlabel("Count")
plt.ylabel("Country")

plt.show()
```



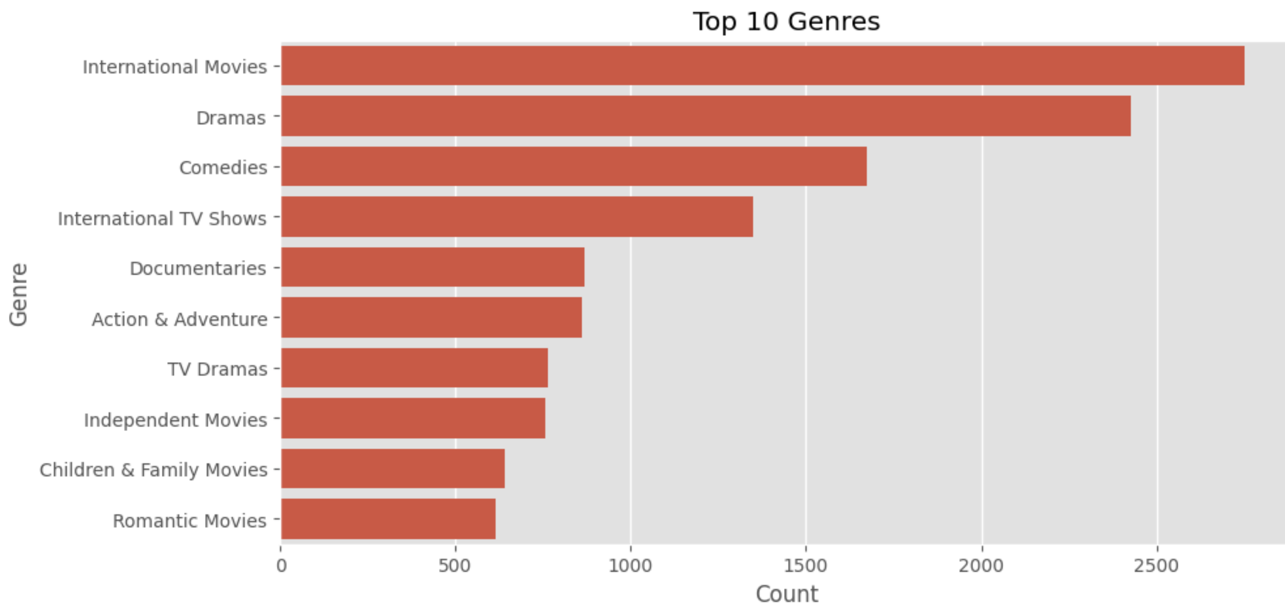
Observation

The United States has the highest number of Netflix titles, followed by India and the United Kingdom.

Step 2: Top 10 Genres Code

```
genre_df = ( df.assign(listed_in=df['listed_in'].str.split(', '))
               .explode('listed_in')
               .reset_index(drop=True) )

plt.figure(figsize=(10,5))

sns.countplot(
    data=genre_df,
```

```
    y='listed_in',
    order=genre_df['listed_in'].value_counts().head(10).index
    )

plt.title("Top 10 Genres")
plt.xlabel("Count")
plt.ylabel("Genre")

plt.show()
```

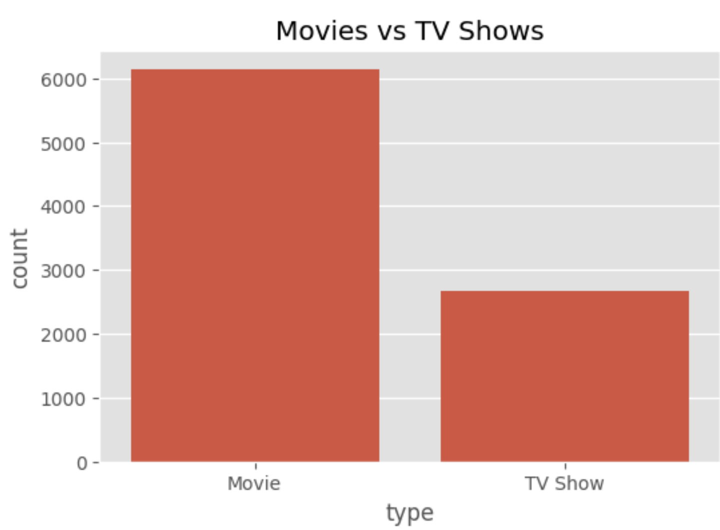


Observation

International Movies is the most common genre available on Netflix.

Step 3: Movies vs TV Shows Code bold text

```
plt.figure(figsize=(6,4))

sns.countplot(data=df, x='type')

plt.title("Movies vs TV Shows")

plt.show()
```

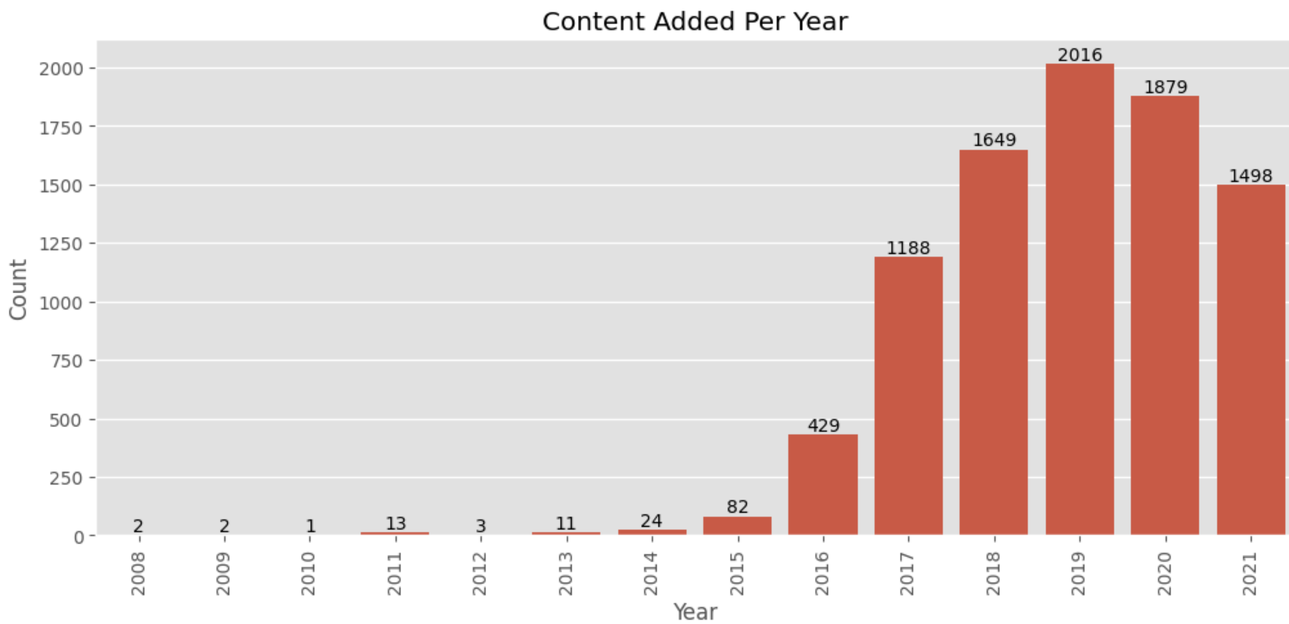


Observation

Netflix offers significantly more Movies than TV Shows.

Step 4: Content Added Per Year Code

```
plt.figure(figsize=(12,5))

ax = sns.countplot(
    data=df,
    x='added_year',
    order=sorted(df['added_year'].dropna().unique())
    )

ax.bar_label(ax.containers[0])

plt.xticks(rotation=90)

plt.title("Content Added Per Year")
plt.xlabel("Year")
plt.ylabel("Count")

plt.show()
```



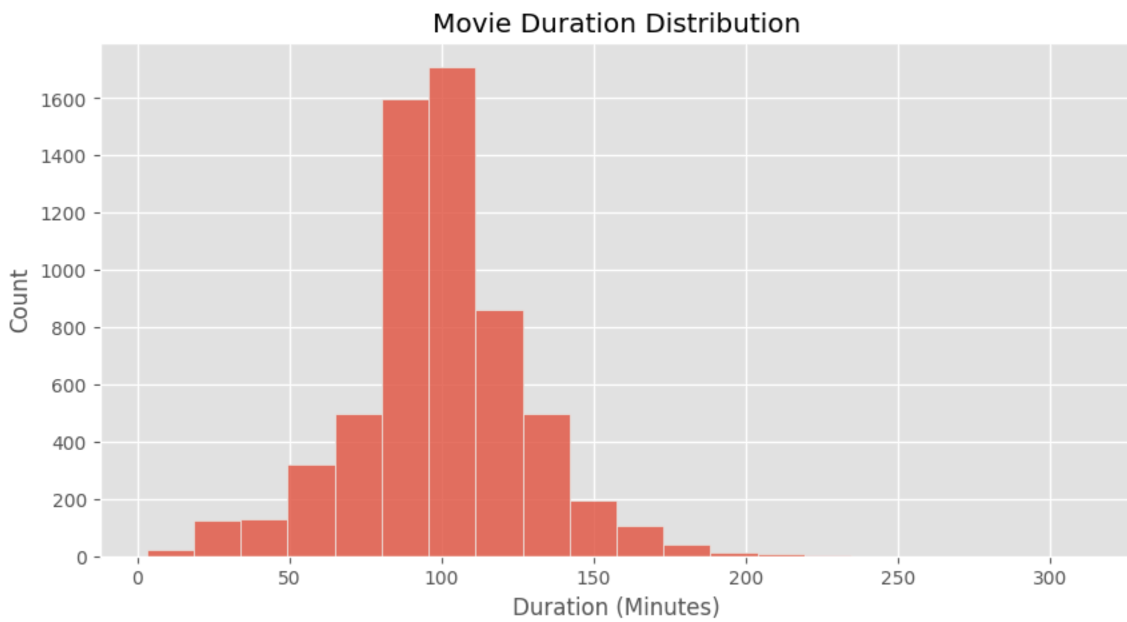
Observation

Netflix added 2,016 titles in 2019, the highest among all years.

Step 5: Movie Duration Distribution Cleaning

```
movies = df[df['type'] == 'Movie'].copy()

movies = movies.dropna(subset=['duration'])

movies['duration'] = (
    movies['duration'].str.replace(' min', '', regex=False).astype(int)
)

plt.figure(figsize=(10,5))
```

```
sns.histplot( data=movies, x='duration', bins=20 )

plt.title("Movie Duration Distribution")
plt.xlabel("Duration (Minutes)")
plt.ylabel("Count")

plt.show()
```

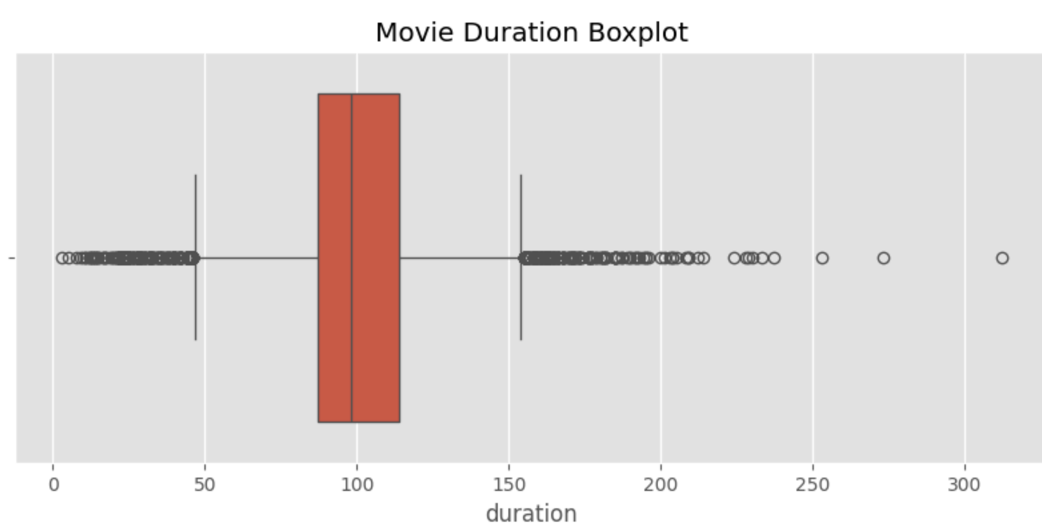


Observation

Most movies have a duration between 80 and 120 minutes.

Step 6: Boxplot (Detect Outliers) Code

```
plt.figure(figsize=(10,4))

sns.boxplot(x=movies['duration'])

plt.title("Movie Duration Boxplot")

plt.show()
```



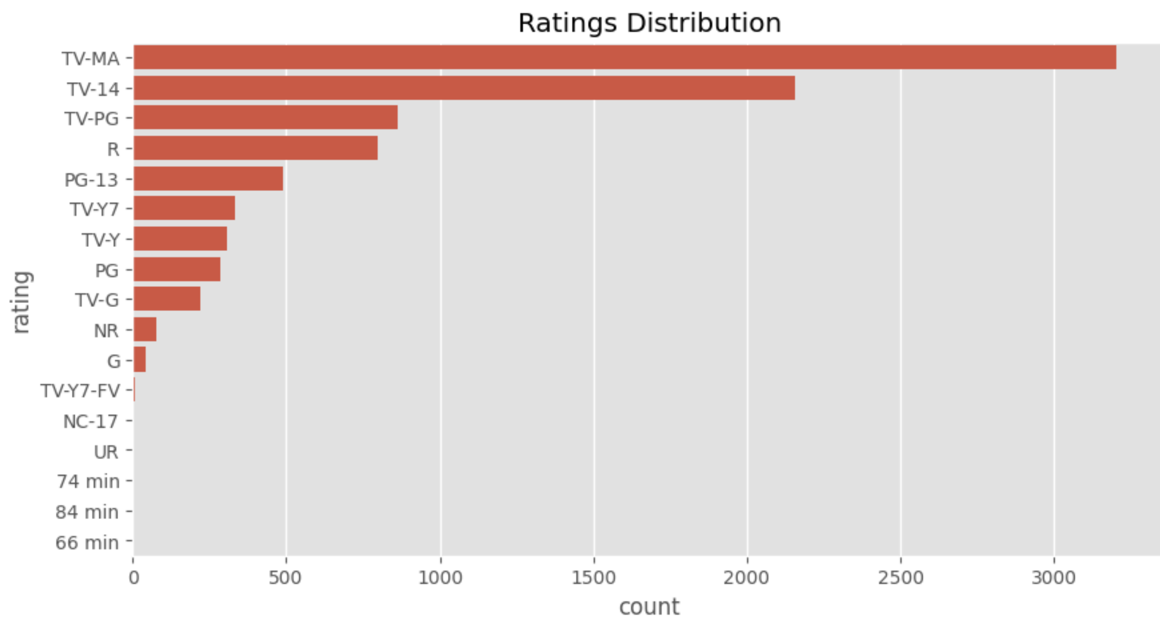
Observation

Most movies are around 100 minutes long, while a few movies longer than 150 minutes appear as outliers.

Double-click (or enter) to edit

Step 7: Ratings Distribution Code

```
plt.figure(figsize=(10,5))

sns.countplot( data=df, y='rating', order=df['rating'].value_counts().index )

plt.title("Ratings Distribution")

plt.show()
```



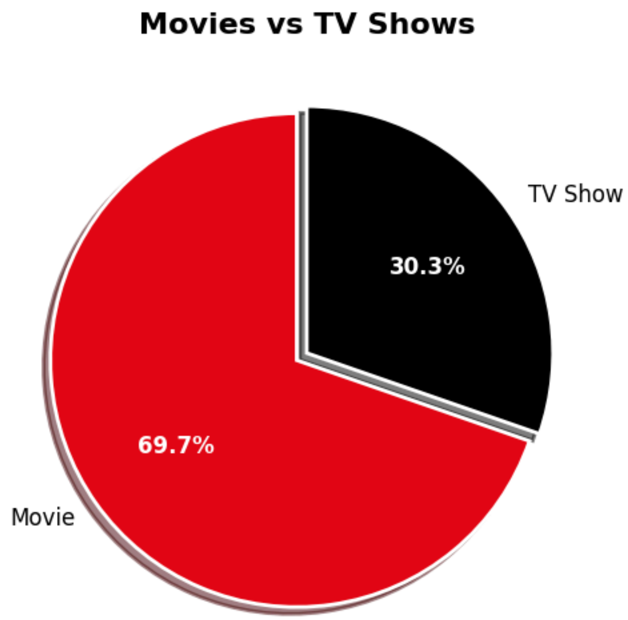
Observation

TV-MA is the most common content rating on Netflix.

Step 8: Movies vs TV Shows (Percentage) Code

```
fig, ax = plt.subplots(figsize=(6,6))

wedges, texts, autotexts = ax.pie(
    df['type'].value_counts(),
    labels=df['type'].value_counts().index,
    autopct='%1.1f%%',
    colors=['#E50914', '#000000'],
    startangle=90,
    explode=(0.05, 0),
    shadow=True,
    wedgeprops={'edgecolor':'white', 'linewidth':2}
)

# Percentage text color
for autotext in autotexts:
    autotext.set_color('white')
    autotext.set_fontsize(12)
    autotext.set_fontweight('bold')
```

```
# Labels color
for text in texts:
    text.set_fontsize(12)

plt.title("Movies vs TV Shows", fontsize=16, fontweight='bold')

plt.show()
```



Observation

Movies account for the majority of Netflix's content library.

Final Business Insights

Business Insight

The United States remains Netflix's largest content supplier, making it the company's most strategic production market.

Netflix's investment in international content indicates strong demand for localized entertainment across global markets.
